information (PII). With that, it could be the case that, akin to sampling, parts of the information to drive correlation are deliberately not present. In addition, privacy legislation or industry compliance standards may demand that signals be handled strictly in-house and not sent to an SaaS provider or cloud provider outside a certain region.

11.4.3 User experience

Last, but most certainly not least, we will discuss challenges related to user experience (UX).

LACK OF ACTIONABLE INSIGHTS

Systems may implement some support for correlation, but I've found that it can be cumbersome or unintuitive to use. In that case, folks tend to not bother and, as a result, not use the feature.

The following is an appropriate remediation: besides (as a vendor) paying attention to UX, as an end user, you could evaluate offerings according to the value of insights they generate. Dynatrace and New Relic both provide (ML-generated) recommendations and insights, which may serve as a starting point for your evaluation.

END-TO-END SUPPORT

Sometimes, the UX challenges come from a lack of end-to-end support. This challenge may affect instrumentation, agents, backends, frontends, and so on.

The following are appropriate remediations:

- Use fully vertical integrated offerings, such as Datadog, New Relic, and Splunk, that own the entire stack.
- Bet on open standards, specifically, OpenTelemetry. At the time of writing, there is one area not yet covered by standards: querying backends. The community has identified this issue and is working on query standardization (http://mng.bz/V1jx) in the context of the CNCF Technical Advisory Group (TAG) on Observability.

11.5 Conclusion

With these considerations on implementing signal correlation in mind, we now conclude not only this chapter but the book itself. Stick around for the appendix, where we will get our hands dirty implementing observability in the context of an multisignal, multiprogramming languages app, using OpenTelemetry.

I would love to hear from you about your challenges and success stories regarding observability and anything else you'd like to share. Please reach out at mh9@o11y.engineering. May the insights be with you!

Summary

- One signal type is usually not enough to gain actionable insights.
- Signal correlation is the metadata-driven process of connecting different signal types to gain actionable insights faster and more accurately.